

DynaMoE: Dynamic Target Interception via Velocity-Conditioned Mixture of Experts

Guozhi Zhan, Bo Wang, Gaoqi Dong, Tianxing Shi, Haomin Wei, Yan Ding, Jun Shang, and Qinyuan Liu

Abstract—While end-to-end imitation learning excels in vision-based robotic manipulation, deploying these policies in highly dynamic environments typically results in severe performance degradation. This failure stems from a fundamental conflict: monolithic networks struggle to internalize mutually exclusive kinematic strategies for intercepting targets at varying velocities while relying on zero-order positional constraints that ignore temporal dynamics. To break this single-speed limitation, we propose DynaMoE, a novel velocity-adaptive IL framework for multi-speed dynamic grasping. By conditioning on an explicit, user-specified velocity prior, DynaMoE introduces a kinematic-guided expert routing mechanism that dynamically allocates computational pathways, achieving physical-level policy decoupling with minimal inference latency overhead. Furthermore, we design an asymmetric spatiotemporal regularization strategy. It applies an exponentially scaled, kinematic-adaptive spatial weighting to tighten interception precision at high speeds and aligns predicted action derivatives with the expert’s reference dynamics through a first-order temporal objective. Evaluated on a newly collected large-scale dynamic grasping dataset using the FastUMI gripper (10,000 trajectories across 40 objects), DynaMoE significantly outperforms state-of-the-art baselines and demonstrates remarkable zero-shot generalization to unseen intermediate velocities. Dataset and code will be made publicly available.

Index Terms—Imitation Learning, Deep Learning in Grasping and Manipulation.

I. INTRODUCTION

END-TO-END imitation learning (IL) has recently demonstrated substantial efficacy in vision-based robotic manipulation, enabling autonomous agents to acquire complex sensorimotor skills directly from human demonstrations [1]–[5]. Conventional approaches primarily focus on static environments, such as table-top pick-and-place or assembly tasks, where objects remain stationary during task execution [6], [7]. To further enhance robustness, the paradigm of action chunking has been introduced, which significantly mitigates compounding errors in long-horizon tasks by predicting future action sequences within a single inference step [8]–[11]. Despite the increasing representational capacity of modern architectures, most sequence-based IL frameworks implicitly

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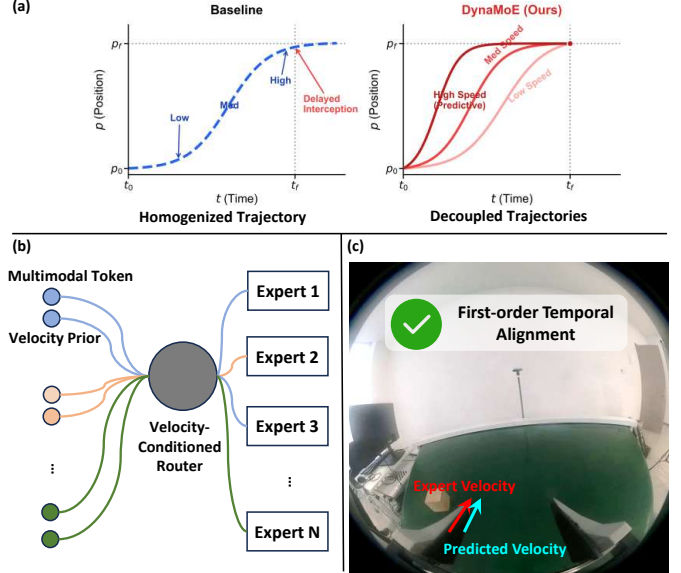


Fig. 1. Overview of dynamic target interception challenges and our proposed DynaMoE framework. (a) Kinematic trajectory analysis. Standard baselines collapse into a single homogenized trajectory (dashed line) for all target speeds, failing to reach the target position p_f by the deadline t_f . In contrast, DynaMoE achieves physical-level policy decoupling. It generates distinct, velocity-specialized trajectories, ensuring precise spatiotemporal convergence. (b) Kinematic-guided routing mechanism. The router dynamically allocates specialized experts by fusing multimodal token representations with the velocity prior. (c) Real-world execution demonstrating precise kinematic consistency and first-order temporal alignment.

remain tethered to a single-speed assumption. Consequently, their performance severely degrades in dynamic scenarios, such as intercepting targets at varying velocities. Although such tasks are ubiquitous in modern industrial automation and logistics, they remain formidable due to the evolution of the environmental state [12]–[14].

Dynamic target interception exposes two fundamental bottlenecks that standard action chunking policies fail to address. First, intercepting objects across varying velocities requires mutually exclusive kinematic strategies: low-speed precision tracking versus high-speed predictive acceleration. Forcing a monolithic network to span these conflicting distributions inevitably induces severe policy interference. At the optimization level, gradients from distinct dynamic regimes destructively interfere, causing the learned action manifold to collapse into a homogenized sub-optimal average [15]. As illustrated in Fig. 1(a), this structural bottleneck prevents the policy from specializing, manifesting as spatial jitters during low-speed execution and unrecoverable temporal lag at high speeds.

Second, conventional architectures optimize exclusively over zero-order positional objectives. This formulation treats each time step’s action as an isolated coordinate, making the policy mathematically blind to first-order velocity and acceleration dynamics. Consequently, the learned manifold captures spatial waypoints, but lacks any explicit representation of temporal execution patterns. They apply uniform spatial penalties regardless of the robot’s dynamic state, depriving the network of explicit gradient guidance to correct temporal phase shifts. Consequently, even if the spatial manifold is perfectly memorized, minor unpunished temporal lags translate to catastrophic failures at high speeds. This absence of temporal grounding forces the model to overfit to static geometries while remaining insensitive to dynamic transients [16], [17].

To address these bottlenecks, we propose DynaMoE, a velocity-adaptive IL framework tailored for multi-speed dynamic interception. Unlike approaches relying on semantic labels or implicit temporal inference, DynaMoE achieves physical-level policy decoupling via a velocity-conditioned Mixture of Experts (MoE) architecture (Fig. 1(b)) [18], [19]. The gating network fuses multimodal tokens with a user-specified velocity prior, dynamically routing the computational graph to specialized experts. This ensures low-speed precision and high-speed aggressiveness with minimal inference latency overhead. Furthermore, to establish explicit temporal grounding, we formulate an asymmetric spatiotemporal regularization strategy. It employs a kinematic-adaptive weighting mechanism to exponentially tighten positional constraints at high speeds, alongside a first-order temporal alignment objective that maps predicted action derivatives to the expert’s reference dynamics (Fig. 1(c)). This dual approach prevents high-frequency overfitting and guarantees correct spatiotemporal synchronization.

To evaluate our framework, we collect a comprehensive dynamic grasping dataset using the FastUMI gripper [20]. In summary, our contributions are four-fold:

- 1) We propose DynaMoE, a velocity-conditioned IL framework that breaks the single-speed limitation of existing action chunking methods, enabling robust interception across a wide range of target velocities.
- 2) We introduce a kinematic-guided expert routing mechanism that dynamically allocates specialized experts by fusing multimodal representations with explicit velocity priors, effectively achieving physical-level policy decoupling to mitigate policy interference.
- 3) We formulate an asymmetric spatiotemporal regularization strategy. By combining kinematic-adaptive spatial weighting with first-order temporal alignment, our strategy explicitly aligns predicted action derivatives with the expert’s action dynamics, accelerating convergence and enhancing high-speed accuracy.
- 4) We construct a large-scale dynamic grasping dataset utilizing the FastUMI gripper, comprising 10,000 demonstration trajectories across 40 diverse objects. Extensive real-world evaluations demonstrate that DynaMoE significantly outperforms state-of-the-art baselines. Dataset and code will be made publicly available soon.

II. RELATED WORK

A. Imitation Learning and Action Chunking

End-to-end IL has emerged as a prominent paradigm for acquiring complex sensorimotor skills directly from expert demonstrations [2], [3]. To mitigate compounding errors, recent advances have shifted from step-by-step regression to sequence-based prediction [1]. Notably, Action Chunking with Transformers (ACT) [8], [9] and Diffusion Policies (DP) series [10], [21] have established new benchmarks for fine-grained manipulation by predicting multi-modal action sequences over a receding horizon. Furthermore, vision-language-action models [5], [6], [22], [23] have unified these architectures for cross-embodiment control. DynaMoE builds upon this action chunking paradigm, inheriting its multi-step prediction capacity. However, these high-capacity models implicitly assume static or single-speed environments. When deployed in highly dynamic scenarios, they frequently suffer from high inference latency [11], making it difficult to maintain the high-frequency control required to intercept fast-moving targets. To cope with this, previous work [14] proposes a rapid adaptive diffusion policy that integrates separate high-frequency replanning modules. However, this adds architectural complexity and introduces additional inference latency, which is undesirable for real-time industrial deployment. Moreover, due to the lack of explicit temporal grounding, they struggle to generalize robustly across diverse kinematic regimes. DynaMoE bridges this gap by directly integrating physical-level policy decoupling and spatiotemporal regularization into the architecture, enabling efficient dynamic execution.

B. Dynamic Target Interception

Manipulating non-cooperative moving targets requires precise spatiotemporal coordination [13]. Traditional systems cascading visual trackers with analytical planners [24], [25] suffer from compounding estimation errors and hardware latency. Alternatively, reinforcement learning (RL) acquires dynamic policies through simulation [26]. However, these approaches demand highly engineered reward functions and remain notoriously susceptible to the sim-to-real gap when deployed on physical hardware [27].

End-to-end IL bypasses these bottlenecks by learning directly from physical demonstrations. While effective in various domains [12], [28], precise adaptation to diverse target speeds remains a formidable bottleneck. Standard action chunking relies on zero-order positional constraints, causing limited temporal sensitivity. Instead of relying on inference-time compensations such as real-time inpainting [16] or masked execution [17], DynaMoE addresses the limitation during policy formulation. While retaining standard generative regularizations, our asymmetric spatiotemporal regularization integrates a kinematic-adaptive spatial weighting with a first-order temporal alignment objective. This explicit guidance forces the model to internalize the expert’s action temporal dynamics, refining the robot’s action manifold and accelerating convergence.

C. Mixture of Experts in Embodied AI

By enabling scalable capacity via sparse routing [29], MoE is increasingly adopted in robotics to alleviate gradient conflicts [15], enhance domain generalization [30], and abstract specialized multi-task skills [19], [31]. Recently, MoE has been integrated into action chunking and diffusion policies [32] to mitigate policy interference by decoupling policies at the task [33] or phase [34] level. However, their reliance on autonomously inferred semantic identifiers or predicted phase transitions makes them ill-equipped to generalize across continuous kinematic regimes. DynaMoE distinguishes itself by achieving physical-level decoupling. Unlike approaches estimating implicit task labels, our routing mechanism dynamically allocates specialized experts based on a user-specified velocity prior, ensuring precise dynamic interception with minimal inference overhead.

III. METHODOLOGY

This section details the DynaMoE framework, designed to extend the action chunking paradigm across diverse velocity regimes. We begin by formulating the velocity-conditioned IL problem, followed by an exposition of the overall architecture and our two core contributions: a kinematic-guided expert routing mechanism and an asymmetric spatiotemporal regularization strategy.

A. Problem Formulation

We formulate the dynamic target interception task as a velocity-conditioned Partially Observable Markov Decision Process (POMDP). At each discrete time step t , the agent receives an observation $o_t = \{I_t, q_t\}$. Here, I_t denotes a single-view visual observation, and $q_t \in \mathbb{R}^{d_q}$ represents the d_q -dimensional proprioceptive state of the manipulator. The environment is conditioned on a global velocity prior $v \in \{0, 1, \dots, N\}$ that represents discrete dynamic regimes. This prior serves as an overarching task-level anchor rather than denoting the instantaneous per-frame target velocity. Although the actual physical speed of the target naturally fluctuates between static waiting and active interception within a single episode, v remains constant to provide a stable dynamic context. It is manually annotated for the training set and is provided as a user-defined task parameter during inference, reflecting known system variables in typical industrial applications. This explicit formulation effectively decouples the global kinematic context from low-level visuomotor control.

The objective of the policy π_θ , parameterized by neural network weights θ , is to map the current observation and dynamic prior to a sequence of future actions. We define this sequence as an action chunk $\hat{\mathbf{A}}_t = \{\hat{a}_i\}_{i=t}^{t+k-1}$, where $\hat{a}_i \in \mathbb{R}^{d_a}$ specifies the d_a -dimensional target control commands and k denotes the prediction horizon. The policy generation process is formalized as:

$$\hat{\mathbf{A}}_t = \pi_\theta(o_t, v; \theta). \quad (1)$$

During training, the model is optimized to minimize the discrepancy between the predicted action sequence $\hat{\mathbf{A}}_t$ and the expert actions $\mathbf{A}_t^*$ across diverse kinematic regimes.

B. DynaMoE Architecture

As illustrated in Fig. 2, DynaMoE utilizes a transformer-based CVAE architecture to model the multimodal action distribution. During training, an auxiliary CVAE encoder processes the ground-truth expert action chunks $\mathbf{A}_t^*$, the proprioceptive state q_t , and the velocity prior v to predict the parameters (μ, σ) of the latent style distribution. The style variable $z \sim \mathcal{N}(\mu, \sigma^2)$ is then sampled using the reparameterization trick. By explicitly conditioning the encoder on v , the resulting latent style variable z is forced to capture pure intra-regime behavioral variations (e.g., human execution noise or stylistic preferences) rather than encoding the macroscopic task speed, effectively disentangling the latent space. During inference, this encoder is discarded and z is sampled from a standard normal prior $\mathcal{N}(0, I)$.

To break the single-speed limitation of standard action chunking, we introduce an explicit kinematic conditioning mechanism. The discrete velocity prior v is mapped to a high-dimensional speed embedding $\mathbf{e}_v \in \mathbb{R}^{d_{\text{model}}}$ via a learnable embedding dictionary. In the encoding phase, $\mathbf{e}_v$ and the style variable z are prepended to the visual and proprioceptive tokens, serving as a global dynamic anchor. This allows the self-attention mechanism to modulate visual features based on the kinematic context.

Within the transformer decoder, we replace the monolithic feed-forward layers with the proposed kinematic-guided expert routing mechanism. While the transformer encoder maintains a unified representation space, the velocity-conditioned MoE decoder dynamically allocates token-level pathways to specialized expert networks. This shared-perception and decoupled-decision architecture allows the model to decouple control strategies across dynamic regimes. By utilizing sparse routing, DynaMoE scales its representational capacity with minimal inference overhead, preserving the high-frequency control loop required for real-world dynamic interception.

C. Kinematic-Guided Expert Routing

To mitigate policy interference, we propose a kinematic-guided expert routing mechanism. For each token $\mathbf{x}_i$ in the decoder's hidden sequence, the gating network determines the expert allocation by concatenating the token's implicit hidden state with the explicit global speed embedding $\mathbf{e}_v$. The routing logits $\mathbf{z}_i$ are computed as:

$$\mathbf{z}_i = \mathbf{W}_g[\mathbf{x}_i; \mathbf{e}_v], \quad (2)$$

where $\mathbf{W}_g \in \mathbb{R}^{N_E \times 2d_{\text{model}}}$ is a learnable routing weight matrix, N_E is the total number of experts, and $[\cdot; \cdot]$ denotes the concatenation of the d_{model} -dimensional hidden state and the speed embedding.

To ensure training stability and expert load balancing, we follow the noisy top- K routing paradigm [29] and introduce Gaussian noise during the routing process:

$$\hat{z}_{i,j} = z_{i,j} + \epsilon_j, \quad (3)$$

where $z_{i,j}$ is the j -th element of the routing logit vector $\mathbf{z}_i$, and $\epsilon_j \sim \mathcal{N}(0, \sigma_{\text{noise}}^2)$ represents the added Gaussian noise sampled for each expert j to facilitate exploration. Subsequently, the

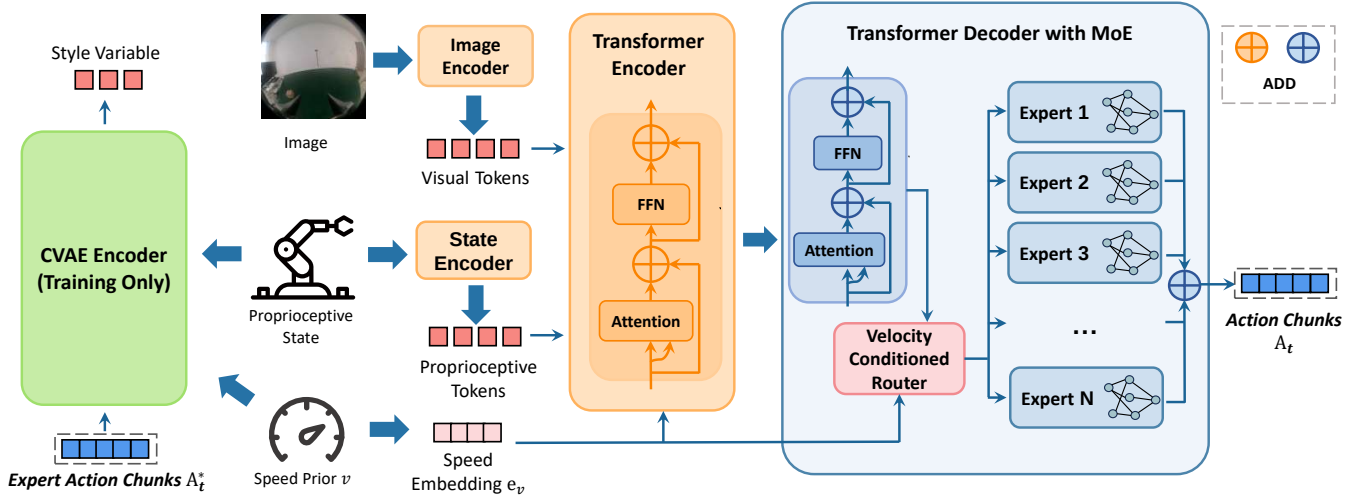


Fig. 2. The overall architecture of DynaMoE. During training, an auxiliary CVAE encoder branch processes expert action chunks, proprioceptive states, and the explicit velocity prior to infer a latent style variable. During policy execution, single-view visual observations and proprioceptive states are encoded and fused with a velocity embedding and the style variable. Within the transformer encoder, this velocity embedding serves as a global kinematic anchor. Within the transformer decoder, a kinematic-guided MoE router dynamically allocates tokens to specialized expert networks based on both the hidden states and the velocity prior, achieving physical-level policy decoupling across diverse dynamic regimes.

global routing probability for the j -th expert is computed via a softmax over all N_E available experts:

$$P_{i,j} = \frac{\exp(\hat{z}_{i,j})}{\sum_{m=1}^{N_E} \exp(\hat{z}_{i,m})}. \quad (4)$$

To enforce sparse activation, we select the top K experts, denoted by the index set $\mathcal{T}_i$ corresponding to the highest K values in the noisy logit vector $\hat{\mathbf{z}}_i$. The final routing weights are then re-normalized among the selected experts:

$$p_{i,j} = \begin{cases} \frac{P_{i,j}}{\sum_{m \in \mathcal{T}_i} P_{i,m}} & \text{if } j \in \mathcal{T}_i, \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

The final output of the MoE component for the token $\mathbf{x}_i$ is the re-weighted sum of the selected experts:

$$\mathbf{y}_i = \sum_{j \in \mathcal{T}_i} p_{i,j} \mathcal{E}_j(\mathbf{x}_i), \quad (6)$$

where $\mathcal{E}_j(\cdot)$ represents the transformation applied by the j -th expert architecture, and $\mathbf{y}_i \in \mathbb{R}^{d_{\text{model}}}$ is the resulting token representation. By guiding routing decisions with the velocity prior $\mathbf{e}_v$, this mechanism achieves physical-level policy decoupling. It enables specific experts to specialize in high-speed interceptions while others maintain spatial precision in low-speed regimes.

D. Asymmetric Spatiotemporal Regularization

Standard action chunking policies optimized solely with zero-order positional constraints struggle to capture temporal kinematics across diverse dynamic regimes. To address this, we introduce a first-order temporal alignment objective $\mathcal{L}_{vel}$, to penalize discrepancies in predicted action derivatives. This finite difference constraint explicitly forces the policy to internalize the expert's action dynamics. For a predicted action

chunk $\hat{\mathbf{A}}_t = \{\hat{a}_i\}_{i=t}^{t+k-1}$, the base differential kinematic loss is defined as:

$$\mathcal{L}_{vel} = \sum_{i=t}^{t+k-2} \| (a_{i+1}^* - a_i^*) - (\hat{a}_{i+1} - \hat{a}_i) \|_1, \quad (7)$$

where a_i^* and $\hat{a}_i$ denote the expert and predicted actions at the i -th step within the prediction horizon.

As the target's speed increases, even minor latency or prediction errors can compound into severe task failures. To enforce stricter alignment in high-speed scenarios, we introduce a velocity-dependent exponential weight. However, applying an absolute exponential penalty can lead to gradient instability when a training batch contains a disproportionate number of high-speed samples. Therefore, we propose a kinematic-adaptive spatial weighting mechanism:

$$\tilde{w}(v_i) = \frac{\alpha^{v_i}}{\frac{1}{B} \sum_{j=1}^B \alpha^{v_j} + \epsilon}, \quad (8)$$

where $v_i \in \{0, 1, 2, \dots\}$ is the discrete velocity prior for the i -th sample in a mini-batch of size B , $\alpha > 1$ is a predefined base factor (empirically set to 1.2 in our experiments to provide bounded exponential scaling), and ϵ is a small stabilizing constant. This context-aware normalization ensures that the optimizer dynamically allocates more gradient focus to higher-speed samples, while strictly preserving the global expected loss scale to ensure stable convergence.

The total training objective combines these terms through the asymmetric spatiotemporal regularization strategy:

$$\mathcal{L}_{total} = \tilde{w}(v) \mathcal{L}_{L1} + \lambda_{vel} \mathcal{L}_{vel} + \lambda_{KL} \mathcal{L}_{KL} + \lambda_{aux} \mathcal{L}_{aux}, \quad (9)$$

where $\mathcal{L}_{L1} = \|\mathbf{A}_t^* - \hat{\mathbf{A}}_t\|_1$ ensures zero-order positional accuracy, $\mathcal{L}_{KL}$ regularizes the style distribution, and $\mathcal{L}_{aux}$ is the auxiliary load-balancing loss ensuring expert utilization efficiency [35]. This asymmetric design decouples the spatial and temporal constraints: the spatial accuracy requirement $\mathcal{L}_{L1}$



Fig. 3. A gallery of the 40 diverse objects utilized in our dynamic grasping dataset. The selected items cover a wide spectrum of shapes, scales, and physical properties, serving to comprehensively evaluate the generalization capabilities of the learned policy across diverse kinematic regimes.

is dynamically tightened via the kinematic-adaptive spatial weighting mechanism $\tilde{w}(v)$ as task speed increases, reflecting the diminishing spatial error tolerance in rapid interceptions. Conversely, the first-order temporal alignment $\mathcal{L}_{vel}$ acts as a universal physical prior for temporal smoothness, controlled by a fixed coefficient λ_{vel} across all dynamic regimes, preventing the network from overfitting to high-frequency temporal noise at extreme speeds.

IV. EXPERIMENTS

The empirical evaluation is designed to answer the following research questions:

- 1) How does DynaMoE compare against state-of-the-art IL baselines across the full dynamic spectrum, and can it generalize to unseen OOD intermediate velocities?
- 2) What are the individual contributions of the kinematic-guided MoE and the asymmetric spatiotemporal regularization in mitigating policy interference and temporal lag?
- 3) Does the velocity-conditioned gating network achieve meaningful, physical-level policy decoupling without suffering from routing collapse?
- 4) To what extent does the first-order temporal alignment objective break the spatial optimization bottleneck and accelerate training convergence?

A. Experimental Setup and Implementation Details

We evaluate DynaMoE on a multi-speed conveyor belt interception task directly deployed on real-world hardware. Demonstrations are directly collected using the FastUMI gripper [20], which inherently relies on a fisheye camera. To eliminate visual discrepancies during deployment, the FastTouch 6-DoF robotic arm used for execution is equipped with an identical wrist-mounted fisheye camera.

In this work, we introduce a large-scale dynamic grasping dataset comprising 10,000 high-quality human demonstrations

across 40 diverse objects, as shown in Fig. 3. The dataset continuously covers five distinct, ascending velocity levels, yielding 50 trajectories per object for each level. However, to rigorously evaluate the zero-shot interpolation capabilities of our framework during physical deployment, we intentionally restrict the training phase to only three discrete velocity anchors: the lowest, medium and highest levels. These anchor regimes—denoted as *Low*, *Medium*, and *High*—are mapped to a progressive velocity prior $v \in \{0, 1, 2\}$ for policy conditioning. The remaining two intermediate velocity levels are strictly excluded from the training data. All our evaluations are conducted strictly through physical real-world trials. Within this physical testing protocol, the excluded intermediate speeds serve purely as OOD targets. This forces the policy to rely on its internalized first-order kinematics and closed-loop visual feedback to dynamically correct and intercept targets at unseen velocities. During physical deployment and inference, this velocity prior v is provided by the user to dictate the desired dynamic behavior.

Our proposed velocity-conditioned MoE architecture allocates token processing across $N_E = 24$ independent expert networks. The router utilizes a noisy top- K gating mechanism where $K = 4$, which balances computational sparsity with collaborative expert representation. For the kinematic-adaptive spatial weighting mechanism introduced in Section III-D, we empirically set the base factor $\alpha = 1.2$. This ensures the spatial penalty is amplified by 1.2^v , enforcing stricter precision at higher regimes. Following the standard action chunking paradigm, all evaluated models predict a future action sequence of $k = 50$ steps at each inference cycle. Operating at a low-level control frequency of 30 Hz, the robotic system executes the first 20 steps of the predicted chunk before querying the policy for the next prediction.

B. Baseline Methods

We compare DynaMoE against three representative baselines, encompassing standard action chunking, task-level mixture-of-experts, and phase-level mixture-of-experts:

- **ACT** [8]: The standard action chunking with transformers architecture, serving as the foundational zero-order spatial baseline.
- **MAE-ACT** [33]: A multi-task architecture that dynamically mixes action expert embeddings based on a predicted task prototype. This baseline represents state-of-the-art advancements in *task-level* policy decoupling.
- **MoE-ACT** [34]: A supervised MoE framework originally designed for phase-structured surgical tasks. By routing tokens based on discrete temporal stages, it represents *phase-level* expert routing paradigms.

Since the official implementations of MAE-ACT and MoE-ACT are not publicly available, we re-implemented their models based on the details provided in their respective publications. For a fair comparison in our dynamic grasping setup, we adapted their gating mechanisms by utilizing the available multimodal observations to infer their original task or phase prototypes.

C. Dynamic Generalization and Interpolation

We evaluate the grasping success rate across objects in our dataset. For each of the five kinematic regimes, we conduct 10 independent physical trials per object, resulting in detailed evaluation of 400 trials per velocity level. A trial is recorded as a success only if the manipulator securely grasps, lifts, and successfully deposits the target into a designated receptacle. As defined in Section IV-A, all models are trained identically on the discrete anchors *Low*, *Medium*, and *High*, while *Low-Med* and *Med-High* serve as OOD velocity regimes.

TABLE I
GRASPING SUCCESS RATES ACROSS TRAINED AND UNSEEN DYNAMIC REGIMES.

Method	Low	Low-Med (OOD)	Medium	Med-High (OOD)	High
ACT [8]	67.7%	42.3%	38.3%	0%	0%
MAE-ACT [33]	74.5%	42.7%	42.4%	5.7%	0%
MoE-ACT [34]	78%	28.6%	8.4%	0%	0%
DynaMoE (Ours)	92.4%	96.8%	97.8%	94.5%	88.2%

The results in Table I reveal significant limitations in existing IL paradigms when confronted with diverse kinematic regimes. Although ACT achieves a moderate success rate of 67.7% at the *Low* anchor speed, its performance degrades as the target speed increases, failing entirely in the *Med-High* and *High* regimes. This failure stems from two fundamental limitations: first, static zero-order positional constraints are insufficient for rapid dynamic interceptions; second, forcing a monolithic network to model diverse kinematic regimes induces severe policy interference, causing the learned action manifold to collapse into homogenized, sub-optimal trajectories that intrinsically lag behind high-speed targets. Furthermore, while task-level MAE-ACT maintains moderate performance at the *Medium* anchor, phase-level MoE-ACT collapses significantly. Both architectures experience severe degradation across OOD interpolation ranges and fail entirely at the *High* regime. This performance gap exposes the fundamental limitations of applying classification-based routing mechanisms to dynamic physical environments. Inherently designed for disjoint multi-task learning, MAE-ACT inappropriately isolates dynamic control into disjoint linear action heads, structurally eliminating interpolative capacity. In contrast, MoE-ACT attempts to blend policies by averaging spatial trajectories across discrete phases, a mechanism that physically corrupts kinematic execution. Moreover, lacking explicit temporal grounding, both architectures rely purely on static spatial memorization. During OOD execution, this structural inflexibility induces token misallocation and temporal lag, causing the manipulator to miss the moving target.

DynaMoE maintains high robustness across the dynamic spectrum, achieving 96.8% in Low-Med and 94.5% in Med-High. This generalization to unseen velocities confirms the efficacy of combining a conditional velocity prior and MoE-based policy decoupling with explicit temporal regularization. While the discrete prior serves purely as a latent anchor that triggers specialized expert pathways, the first-order temporal objective $\mathcal{L}_{vel}$ grounds these embeddings to physical kinematics. By penalizing action derivatives across these decoupled pathways, the policy correlates the conditional prior with

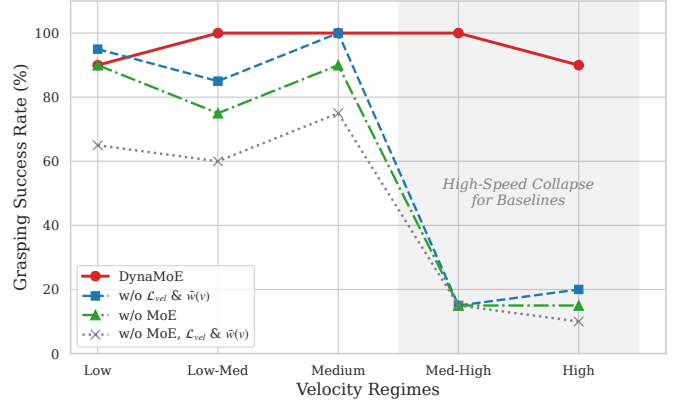


Fig. 4. Ablation study on grasping success rates across different velocity regimes. DynaMoE outperforms all ablated variants, successfully preventing the performance collapse across both trained velocity anchors and unseen intermediate speeds.

temporal dynamics. Rather than memorizing discrete spatial waypoints, DynaMoE internalizes the underlying first-order kinematics, enabling optimal interpolation across varying velocity domains without suffering from policy interference. Crucially, DynaMoE achieves this kinematic generalization using only three training anchors and a mere 50 demonstration trajectories per object at each velocity. This highlights the data efficiency of our framework, proving that explicit spatiotemporal regularization significantly reduces the sample complexity typically required for continuous dynamic control.

Despite this robust generalization, DynaMoE exhibits a performance drop within the *High* kinematic regime. We attribute these remaining failures primarily to the critically narrow temporal window for interception. As the target rapidly traverses the workspace, the duration it remains graspable decreases drastically, imposing a low tolerance for predictive variance. Consequently, even minor temporal lag or spatial misalignments, which are typically compensated by closed-loop visual feedback at lower speeds, become unrecoverable and lead to terminal grasping failures. This tight temporal constraint is further compounded by sensory degradation. Fixed-exposure fisheye cameras suffer from visual motion blur during the high-speed approach, occasionally corrupting the spatial features and exacerbating the predictive variance.

D. Ablation Study

To quantify the contribution of each architectural component, we conducted a leave-one-out ablation study, detailed in Fig. 4 across both trained anchor speeds and OOD interpolation intervals. DynaMoE consistently outperforms all ablated variants, preventing the performance collapse observed in baselines during high-speed and unseen dynamic interceptions.

First, removing the asymmetric spatiotemporal regularization induces severe performance degradation, falling to a 15% success rate in the unseen *Med-High* regime despite adequate performance at the trained *Medium* speed. Optimizing exclusively with zero-order positional constraints deprives the network of speed awareness. Without the first-order temporal alignment objective, the policy successfully predicts spatial

interception coordinates but fails to synchronize with the expert’s demonstrated execution dynamics, resulting in temporal lag and grasp failures.

Second, discarding the MoE architecture forces a monolithic network to model diverse kinematic regimes. Even with asymmetric spatiotemporal regularization, this variant suffers severe policy interference, dropping to 15% success in the *High* regime and interpolating poorly across OOD ranges. The high joint accelerations required for high-speed interceptions conflict with the precise movements of low-speed tasks. Within a shared parameter space, these conflicting gradients average out, producing homogenized, temporally lagged trajectories.

Finally, removing both MoE routing and temporal regularization yields the worst generalization, dropping to 10%–15% in the *High* and OOD intervals. This substantiates our hypothesis: concatenating a discrete velocity prior to the observation space is insufficient for generalizing across diverse kinematic regimes. Without explicit structural routing, the self-attention mechanism dilutes the sparse velocity prior amidst dense visual tokens; without temporal regularization, the network cannot internalize the expert’s temporal execution.

E. Interpretability of Kinematic-Guided Expert Routing

A common critique of MoE architectures is their black-box nature and the tendency for routing collapse. To demonstrate that DynaMoE achieves meaningful physical-level decoupling, we analyze the activation frequencies of the $N_E = 24$ experts across diverse kinematic regimes.

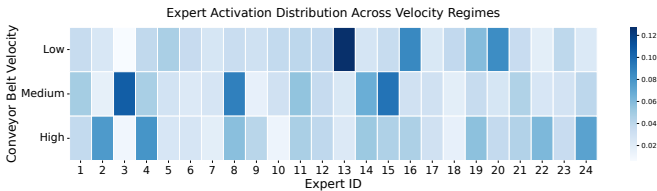


Fig. 5. Heatmap of expert activation probabilities across three velocity regimes. The routing network dynamically shifts its preference based on the environmental velocity, demonstrating distinct expert specialization while maintaining shared pathways for common features.

By visualizing the routing probability distribution in Fig. 5, we observe distinct expert specialization that directly aligns with the underlying physical execution principles. Under the Low kinematic regime, the gating network predominantly routes tokens to a sparse subset of experts (e.g., Experts 13 and 16). This localized activation reflects the quasi-static nature of low-speed tasks, which demand high zero-order spatial precision. Conversely, as velocity increases across the Medium and High domains, the routing distribution broadens to activate a collaborative ensemble of experts (e.g., 2, 3, 4, 8 and 15). Dynamic target interception is a highly coupled spatiotemporal task, requiring simultaneous temporal synchronization and acceleration compensation. Furthermore, the persistent activation of specific experts (e.g., Experts 11 and 14) across all velocity regimes indicates the spontaneous emergence of shared feature extractors. These universally active experts likely encode velocity-invariant representations,

such as fundamental spatial geometry or the manipulator’s proprioceptive state. Ultimately, by segregating conflicting optimization gradients into distinct network compartments while preserving shared invariant pathways, our kinematic-guided MoE mechanism effectively prevents policy interference and manifold collapse that typically paralyze monolithic baselines.

F. Convergence Analysis and Optimization Efficiency

In IL for dynamic tasks, reaching the correct spatial coordinate is intrinsically coupled with the robot’s temporal execution. To quantitatively assess the impact of the proposed first-order temporal alignment objective, we analyze the training efficiency and spatial convergence profiles of DynaMoE with and without this constraint.

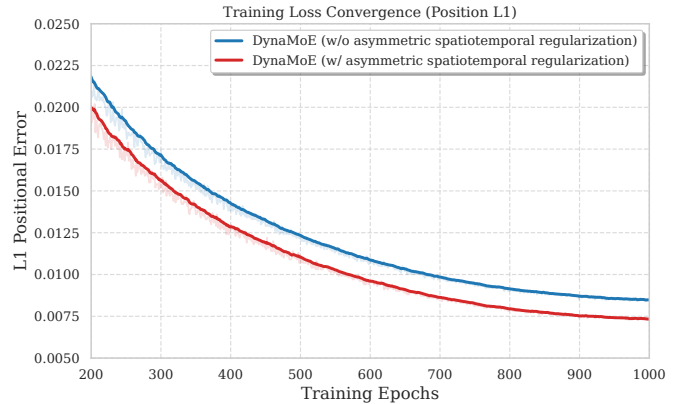


Fig. 6. Training loss convergence profile focusing on the deep optimization phase (Epochs 200 to 1000). DynaMoE equipped with asymmetric spatiotemporal regularization consistently maintains a lower spatial error margin and converges to a distinctly tighter asymptotic bound compared to the variant relying solely on zero-order constraints.

As depicted in Fig. 6, we evaluated the L1 positional error focusing on the mid-to-late training stages. While the variant optimized exclusively with zero-order positional constraints steadily converges, it eventually plateaus at a higher asymptotic error bound. This aligns with our hypothesis that relying solely on positional data limits the network’s ability to fully capture the expert’s action dynamics, resulting in suboptimal spatial precision.

Acting as a physical inductive bias, the temporal derivative constraint helps the policy break through the spatial optimization bottleneck and achieve a tighter convergence bound. This enhanced trajectory precision directly translates to the higher success rates observed in the *High* regime.

V. CONCLUSION

In this work, we present DynaMoE, an imitation learning framework tailored for multi-speed dynamic grasping. To overcome the policy interference inherent in traditional action chunking, we introduce a kinematic-guided expert routing mechanism that achieves physical-level policy decoupling via a global velocity prior. Furthermore, our asymmetric spatiotemporal regularization employs a first-order temporal alignment objective to precisely synchronize the policy’s temporal execution across diverse dynamic regimes. Real-world

evaluations demonstrate that DynaMoE significantly outperforms state-of-the-art baselines. Notably, it achieves zero-shot generalization to unseen out-of-distribution velocities with exceptional data efficiency, relying on limited training anchors and minimal demonstrations, all while maintaining the high-frequency inference essential for physical deployment. To inspire further research into high-agility embodied agents, we open-source both the DynaMoE framework and our dynamic grasping dataset comprising 10,000 high-quality trajectories.

While DynaMoE demonstrates strong generalization across the discrete kinematic regimes considered in this work, several limitations remain avenues for future investigation. Future work will extend this framework to intercept targets with time-varying velocities, such as accelerating or decelerating objects. Furthermore, we aim to leverage representation learning to implicitly estimate continuous kinematic states directly from streaming visual and proprioceptive inputs. This advancement will ultimately eliminate the reliance on manual prior specification and drive the development of fully autonomous open-world dynamic manipulation.

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